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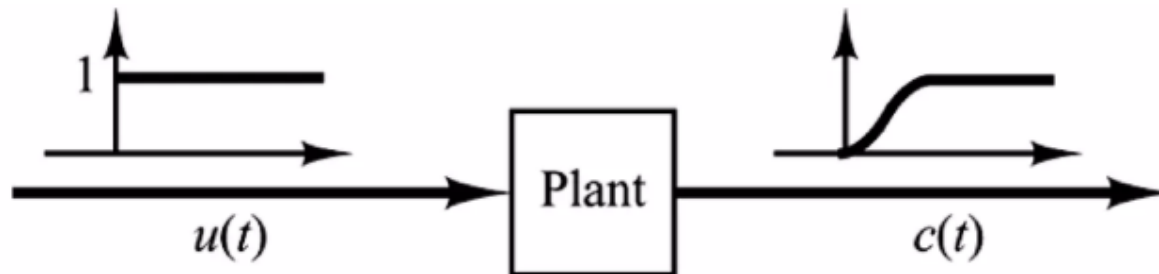
Autonomes Fliegen

WS 2025

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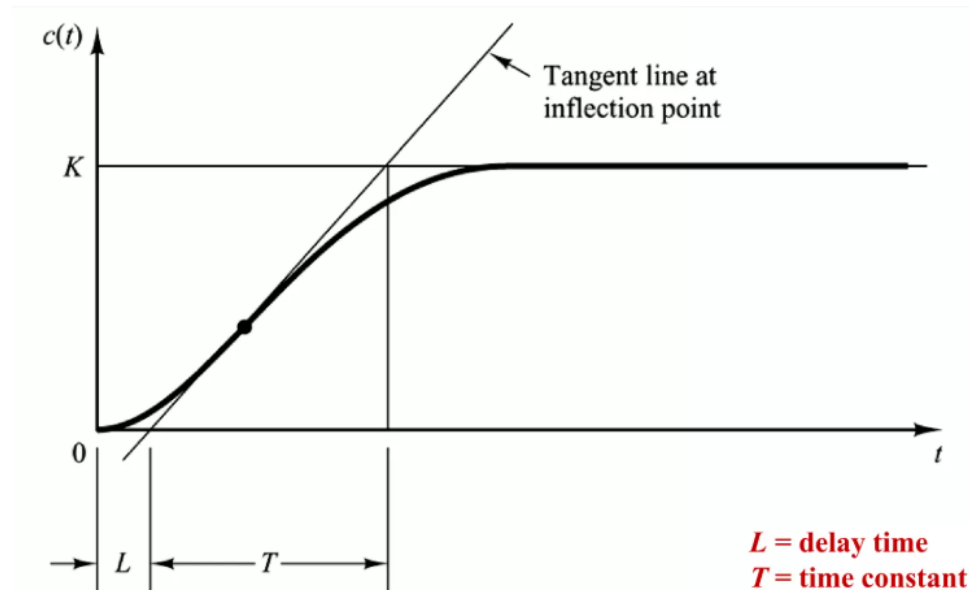
- Step Input method/ Open loop method.
- Critical gain - critical period method/ Close loop method.



Time Response 1

Open-Loop Step Response

- This method captures the plant's response experimentally, used when a step input yields an S-shaped curve.
- The S-curve is defined by Delay Time (L) and Time Constant (T), found by drawing a tangent at the inflection point and noting its intersections with the time axis and output level K .
- Ziegler-Nichols sets controller parameters K_p , T_i , and T_d using formulas in the table.



- The transfer function $\frac{C(s)}{U(s)}$ may then be approximated by a first-order system with a transport lag as follows:

$$\frac{C(s)}{U(s)} = \frac{K e^{-Ls}}{Ts + 1}$$

- Ziegler and Nichols suggested to set the values of K_p , T_i , and T_d according to the formula shown in Tabel 1.

Time Response 3

Open-Loop Step Response

- For which controller: P, PI, or PID controllers.

How it works: This method analyzes the response of the system when a step input is applied while in open-loop mode.

- Process: With the controller in manual mode and the system at a steady state, apply a small step change to the input. Record the resulting process output over time.
- From the resulting step response, determine the system parameters: K: The steady-state gain. L: The dead time (delay time). T: The time constant.

Time Response 4

PID Controller Parameters in the Ziegler-Nichols Step Response Method

Table 1: Ziegler Nichols Tuning Chart

Type of Controller	K_p	T_i	T_d
P	$\frac{T}{L}$	∞	0
PI	$0.9 \frac{T}{L}$	$\frac{L}{0.3}$	0
PID	$1.2 \frac{T}{L}$	$2L$	$0.5L$

Time Response 5

PID Controller Parameters in the Ziegler-Nichols Step Response Method

- Notice that the PID controller tuned by the first method of Ziegler-Nichols rules gives

$$\begin{aligned} G_c(s) &= K_p \left(1 + \frac{1}{T_i s} + T_d s \right) \\ &= 1.2 \frac{T}{L} \left(1 + \frac{1}{2Ls} + 0.5Ls \right) \\ &= 0.6T \frac{\left(s + \frac{1}{L} \right)^2}{s} \end{aligned}$$

- Thus, the PID controller has a pole at the origin and double zeros at

$$s = -\frac{1}{L}$$

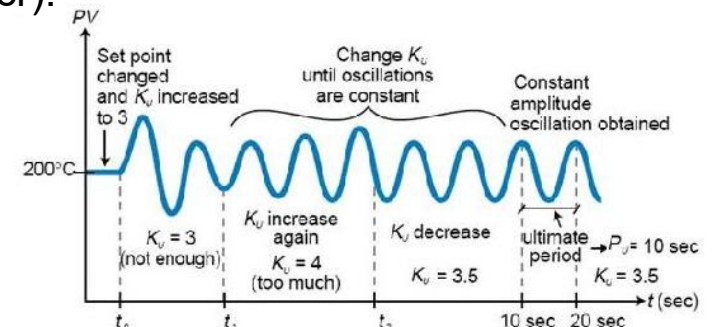
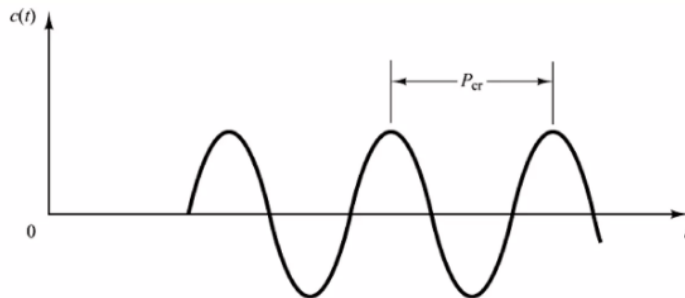
The Ziegler-Nichols tuning method (Second method) 1

Critical Gain – Critical Period

- This method, based on continuous cycling, is performed experimentally on a system that is running in a closed-loop configuration.

How to find K_{cr} and P_{cr} :

1. **Set up the controller.** Put the controller into P-only mode by setting the integral (I) and derivative D gains to zero.
2. **Increase the proportional gain (K_p).** Slowly increase the value of (K_p) from zero. Apply a small disturbance, such as a setpoint change, and observe the system's output.
3. **Identify sustained oscillations.** Continue increasing (K_p) until the output oscillates with a constant, sustained amplitude. This means the oscillations are neither decaying nor growing over time.
4. **Record the ultimate gain (K_{cr}).** The value of (K_p) at which these sustained oscillations occur is the ultimate gain, (K_{cr}).
5. **Measure the ultimate period (P_{cr}).** Measure the period of one full cycle of the sustained oscillation. This is the ultimate period, (P_{cr}).



The Ziegler-Nichols tuning method (Second method) 2

Critical Gain – Critical Period

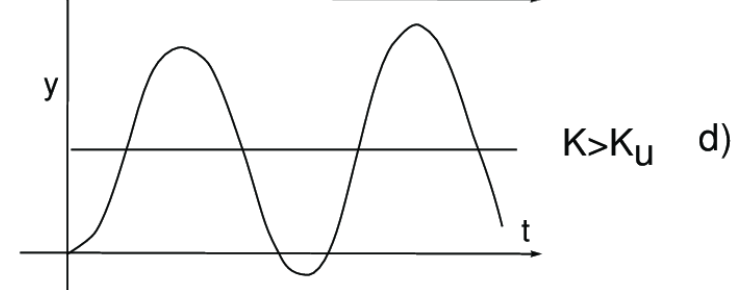
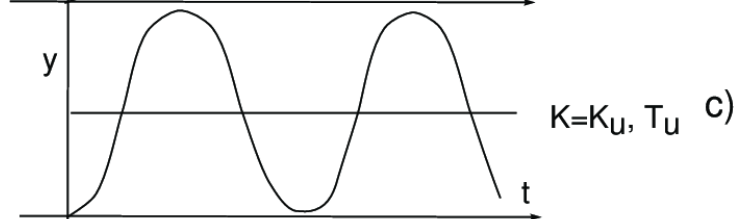
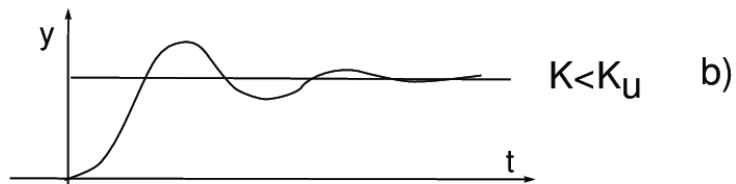
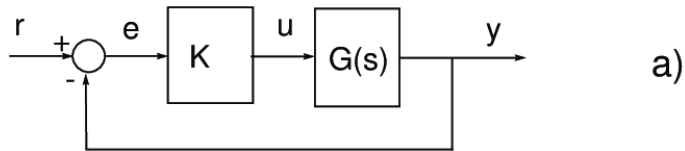


Table 2: Ziegler–Nichols method

Controller	K_p	T_i	T_d
P	$0.5 K_{cr}$	Infinity	0
PI	$0.45 K_{cr}$	$P_{cr}/1.2$	0
PID	$0.6 K_{cr}$	$P_{cr}/2$	$0.125 P_{cr}$

The Ziegler-Nichols tuning method (Second method) 3

Critical Gain – Critical Period

- Notice that the PID controller tuned by the second method of Ziegler-Nichols rules gives

$$\begin{aligned}
 G_c(s) &= K_p \left(1 + \frac{1}{T_i s} + T_d s \right) \\
 &= 0.6 K_{cr} \left(1 + \frac{1}{0.5 P_{cr} s} + 0.125 P_{cr} s \right) \\
 &= 0.075 K_{cr} P_{cr} \frac{\left(s + \frac{4}{P_{cr}} \right)^2}{s}
 \end{aligned}$$

- Thus, the PID controller has a pole at the origin and double zeros at

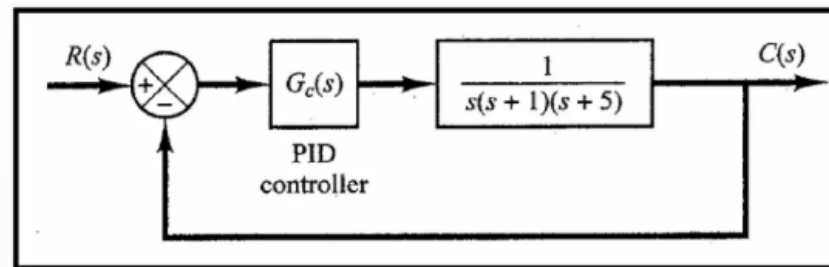
$$s = \frac{-4}{P_{cr}}$$

- Note that if the system has a known mathematical model (such as the transfer function), then we can use the root-locus method to find the critical gain K_{cr} , and the frequency of the sustained oscillations ω_{cr} , where $2\pi/\omega_{cr} = P_{cr}$. These values can be found from the crossing points of the root-locus branches with the $j\omega$ -axis. (Obviously, if the root-locus branches do not cross the $j\omega$ -axis, this method does not apply.)
- Comments. Ziegler-Nichols tuning rules (and other tuning rules presented in the literature) have been widely used to tune PID controllers in process control systems where the plant dynamics are not precisely known. Over many years, such tuning rules proved to be very useful. Ziegler-Nichols tuning rules can, of course, be applied to plants whose dynamics are known. (If the plant dynamics are known, many analytical and graphical approaches to the design of PID controllers are available, in addition to Ziegler-Nichols tuning rules.)

$$G_c(s) = K_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$

- apply a Ziegler-Nichols tuning rule for the determination of the values of parameters K_p , T_i , and T_d . Then obtain a unit-step response curve and check to see if the designed system exhibits approximately 25% maximum overshoot. If the maximum overshoot is excessive (40% or more), make a fine tuning and reduce the amount of the maximum overshoot to approximately 25% or less.

PID-controlled system.



- Since the plant has an integrator, we use the second method of Ziegler-Nichols tuning rules.
- By setting $T_i = \infty$, and $T_d = 0$, we obtain the closed-loop transfer function as follows:

$$\frac{C(s)}{R(s)} = \frac{K_p}{s(s+1)(s+5) + K_p}$$

- The value of K_p , that makes the system marginally stable so that sustained oscillation occurs can be obtained by use of Routh's stability criterion. Since the characteristic equation for the closed-loop system is

$$s^3 + 6s^2 + 5s + K_p = 0$$

- the Routh array becomes as follows:

s^3	1	5
s^2	6	K_p
s^1	$\frac{30 - K_p}{6}$	
s^0	K_p	

- Examining the coefficients of the first column of the Routh table, we find that sustained oscillation will occur if $K_p = 30$. Thus, the critical gain K_{cr} , is

$$K_{cr} = 30$$

- With gain K_p , set equal to K_{cr} ($= 30$), the 2nd-order characteristic equation becomes

$$6s^2 + 30 = 0$$

- To find the frequency of the sustained oscillation, we substitute $s = j\omega$ into this characteristic equation as follows:

$$6(j\omega)^2 + 30 = 0$$

$$6(5 - \omega^2) = 0$$

- from which we find the frequency of the sustained oscillation to be $\omega^2 = 5$ or $\omega = \sqrt{5}$. Hence, the period of sustained oscillation is

$$P_{cr} = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{5}} = 2.8099$$

- Referring to Table 10-2, we determine K_p , T_i , and T_d as follows:

$$K_p = 0.6K_{cr} = 18$$

$$T_i = 0.5P_{cr} = 1.405$$

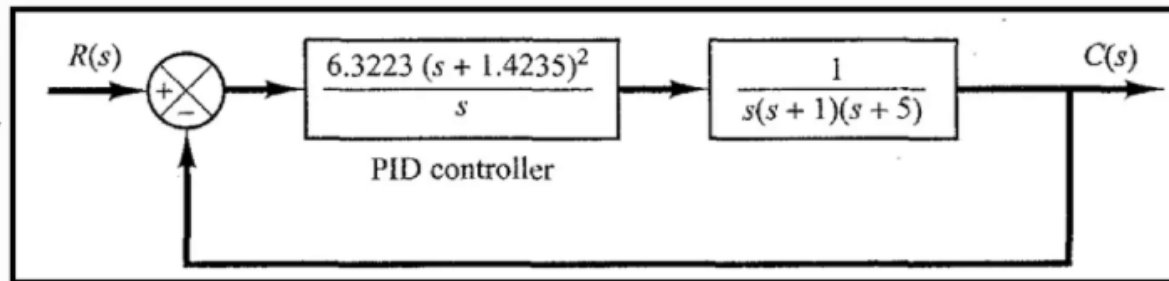
$$T_d = 0.125P_{cr} = 0.35124$$

- The transfer function of the PID controller is thus

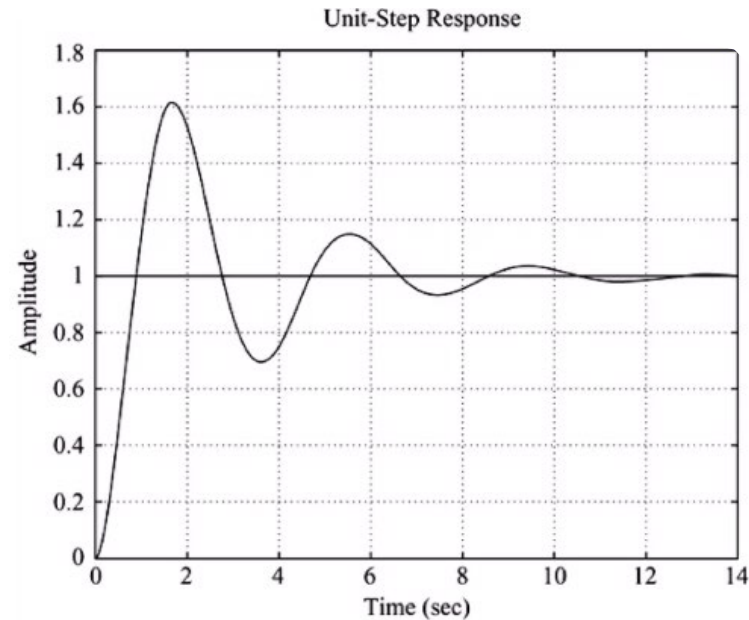
$$\begin{aligned}
 G_c(s) &= K_p \left(1 + \frac{1}{T_i s} + T_d s \right) \\
 &= 18 \left(1 + \frac{1}{1.405s} + 0.35124s \right) \\
 &= \frac{6.3223(s + 1.4235)^2}{s}
 \end{aligned}$$

- The PID controller has a pole at the origin and double zero at $s = -1.4235$. A block diagram of the control system with the designed PID controller is shown in Figure 10-7.

0-7 **Block diagram of the system with PID controller designed by use of the Ziegler–Nichols tuning rule (second method).**



- ❑ The maximum overshoot in the unit-step response is approximately 62%. The amount of maximum overshoot is excessive.



- It can be reduced by fine tuning the controller parameters. Such fine tuning can be made on the computer. We find that by keeping $K_p = 18$ and by moving the double zero of the PID controller to $s = -0.65$, that is, using the PID controller

$$G_c(s) = 18 \left(1 + \frac{1}{3.077s} + 0.7692s \right) = 13.846 \frac{(s + 0.65)^2}{s}$$

- the maximum overshoot in the unit-step response can be reduced to approximately 18%

